```
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import StandardScaler

from sklearn.discriminant_analysis import LinearDiscriminantAnalysis

from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

df = pd.read_csv("Iris.csv")

df .head()


print (df.shape)
print (df.info())

df[['SepalLengthCm', 'SepalWidthCm',
"PetalLengthCm', ‘PetalWidthCm' ]]

y = df['Species']

y-map({
‘Iris-setosa': @,
"Iris-versicolor': 1,
"Iris-virginica’: 2

print(y.head())

X_train, X_test, y_train, y_test = train_test_split(X, Y, test_size=0.2, random_state=42, stratify=y

scaler = StandardScaler()

X_train = scaler. fit_transform(X_train)
X_test = scaler.transform(X_test)

lda = LinearDiscriminantAnalysis(n_components=2)

X_train_lda = lda.fit_transform(X_train, y_train)
X_test_lda = lda.transform(X_test)
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y_pred = lda.predict(X_test)

print(y_pred)


accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)
print("Accuracy: {:.2f}%".format(accuracy * 10@))


print(classification_report(
y_test,
y_pred,
target_names=[
"Iris-setosa’,
"Iris-versicolor',
"Iris-virginica'

))


cm = confusion_matrix(y_test, y_pred)

print(cm)

plt.figure(figsize=(6, 5))

plt.imshow(cm)
plt.colorbar()

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("LDA Confusion Matrix")

plt.show()


plt.figure(figsize=(8, 6))

for i, species in enumerate([
"Iris-setosa',
```

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"Iris-versicolor',
"Iris-virginica'

]):
plt.scatter(
X_train_lda[y_train == i, 0],
X_train_lda[y_train == i, 1],

label=species

)

plt.xlabel("LD1")

plt.ylabel("LD2")

plt.title("LDA Projection of Iris Dataset")
plt.legend()

plt.show()
```